Bayesian Modelling of Component K Degradation

Investigating the reliability of the mechanical component "K"

Executive Summary

The hierarchical Bayesian model of component K integrity presented in this report delivers reliable predictions across the full range of available data, including the 25 K components with sparse data. Two model variants were fitted: the baseline exponential decay model and an enhanced model in which the diagnostics X1–X5 adjust the decay rate. Both models were fitted with the Hamiltonian Monte Carlo (HMC) sampler and "No U-Turn" (NUTS) kernel.

Three diagnostics emerge as **strong predictors of degradation**: **X3** (largest effect with standardised mean weight +1.50), **X4** (+0.67) and **X2** (+0.41), all with 95% credible intervals far from 0 (no effect). X1 and X5 have little to no effect on component integrity. We supply the 25 blind test probabilities in predictions.csv and the self contained, enhanced model predictor in deployable.ipynb .

1. Introduction

The client's challenge is given irregular, noisy and often sparse integrity measurements for 75 K components to produce predictions for future integrity that are both accurate and explain the level of uncertainty.

2. Methodology

2.1 Hierarchical Bayesian Model

Two models were fit: the baseline model and the enhanced model. Both models share the same structure: integrity of component i, y_i, at time t is treated as a exponential decay term plus a measurement noise term,

$$y_i = f_i(t) + \epsilon_i, \quad \epsilon_i \sim \text{Normal}\left(0, \sigma_y^2\right)$$

For the **baseline model**, the decay term is $f_i(t) = u_i \exp(-v_i t/100)$. The intercept u_i and decay rate v_i are specific to component i, drawn from normal distributions:

$$u_i \sim Normal(\mu_u, \sigma_u^2),\ v_i \sim Normal(\mu_v, \sigma_v^2).$$

The hyperpriors for $\mu_u, \sigma_u^2, \mu_v, \sigma_v^2$ are weakly informative and explained in Section 2.3.

In the **enhanced model**, the decay rate is also controlled by a shared linear combination of standardised diagnostics:

$$f_i(t) = u_i \exp\left\{-\left(v_i + \sum_{j=1}^{5} w_j \tilde{x}_{ji}\right) t/100\right\}, \quad w_j \sim \text{Normal}(0, \sigma_w^2), \quad \sigma_w \sim \text{HalfNormal}(1)$$

We standardise the diagnostics $\left(\tilde{x_{ji}} = \left(x_{ji} - \bar{x_j}\right)/s_j\right)$ to mean 0 and standard deviation 1 (by subtracting the mean and dividing by the standard deviation) so that the magnitudes of the weights w_j are comparable across the five features.

2.2 Sampling

The posterior distribution of all unknowns is analytically intractable and so stochastic approximation must be used: Hamiltonian Monte Carlo (HMC) sampler and "No U-Turn" (NUTS) kernel implemented using the numpyro probabilistic programming library. HMC with NUTS is considered state of the art for Markov Chain Monte Carlo approximation/stochastic approximation because it adapts trajectory length automatically and makes use of gradient information to generate independent proposals each time, which makes the algorithm very computationally efficient.

4 parallel chains were ran, each of 2,000 warm-up and 2,000 actual samples, giving 8,000 retained posterior samples in total. The samples are reliable given that r̂ was 1.00 for every variable, effective sample sizes made up more than 10% of total samples for every variable and there were zero divergences. The traces in Figures 1 and 3 look like random noise, meaning that the sampler explored the space efficiently confirming reliability. Divergences were also avoided using reparametrisation when sampling for u and v.

2.3 Choice of Priors

The priors chosen encode our prior knowledge about the K components. For the intercept hyperprior, $\mu_u \sim Normal(90, 10^2)$ was chosen, centred on the midpoint of the client's stated 80 to 100 range and with enough spread to cover the whole region at one standard deviation. For the decay rate, $\mu_v \sim Normal(5, 4^2)$ was chosen, again centred on the client's stated 1 to 10 range; the reasonably large standard deviation of 4 ensures that the whole region is covered and that the prior does not unnecessarily constrain the posterior distribution.

The baseline and enhanced models share priors for all variables present in both models, excluding the decay rate hyperprior. $\mu_v \sim Normal(4, 4^2)$ was chosen instead for the enhanced model: a slightly smaller mean for the normal distribution. This is to account for the v_i being expected to be smaller in magnitude in the enhanced model because of the incorporation of the diagnostics in the decay term.

The half Normal priors for $\sigma_u, \sigma_v, \sigma_y, \sigma_w$ are weakly informative with different scales for their half Normal priors chosen to account for the magnitude of the corresponding u, v, y and w values. These softly constrain the different standard deviation values to plausible ranges.

There was no prior knowledge about the size of the diagnostic weights w_j and so $w_j \sim Normal(0, \sigma_w^2)$ was chosen. This is almost as though assuming that the diagnostics do not influence component K degradation: weights that the data do not strongly support are shrunk to 0.

We can see from Section 4 that this prior also does allow weights that are well-evidenced to deviate from 0: for example, w_3 learns a posterior mean of +1.50.

3. Model Results

3.1 Baseline Model

The trace plot of the baseline model (see Figure 1) shows the posterior distributions of all variables are unimodal and not heavily skewed and that the four chains are largely indistinguishable from each other. The population level posterior means are μ_u = 89.72 (95% CI 88.3–91.1), μ_v = 3.93 (95% CI 3.6–4.3), σ_u = 5.28, σ_v = 1.59 and measurement noise σ_y = 5.10. These are consistent with the client's estimate of $u_i \in [80, 100]$, $v_i \in [1, 10]$) and so the model seems appropriate.

The model's predictions for four different components are shown in Figure 2. For K#0003 (40 observations – lots of data), the posterior mean function follows the data closely and has small uncertainty bands. The three components from the test set (K#0055, K#0065, K#0074) have limited data and so the uncertainty bands are quite large, particularly at areas away from the available data. These three predictions illustrate the value of the hierarchical prior: even with one or four observations, the model gives a sensible mean prediction, borrowing information from the population mean. For K#0055 in particular, the 95% predictive interval at t = 45 spans 0 to nearly 60% and represents the range of likely possible integrities for this component on day 45.

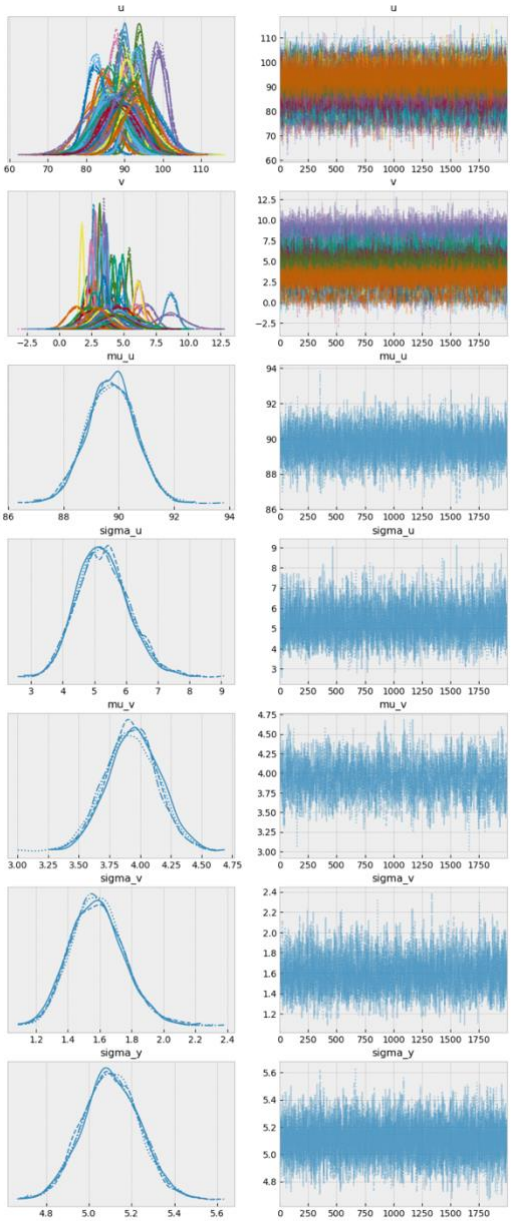


Figure 1. Baseline model posterior diagnostics. Posterior distributions, one curve per chain, and trace of sampled values for component parameters and measurement noise. The first two rows show all 75 per component u_i and v_i on one graph.

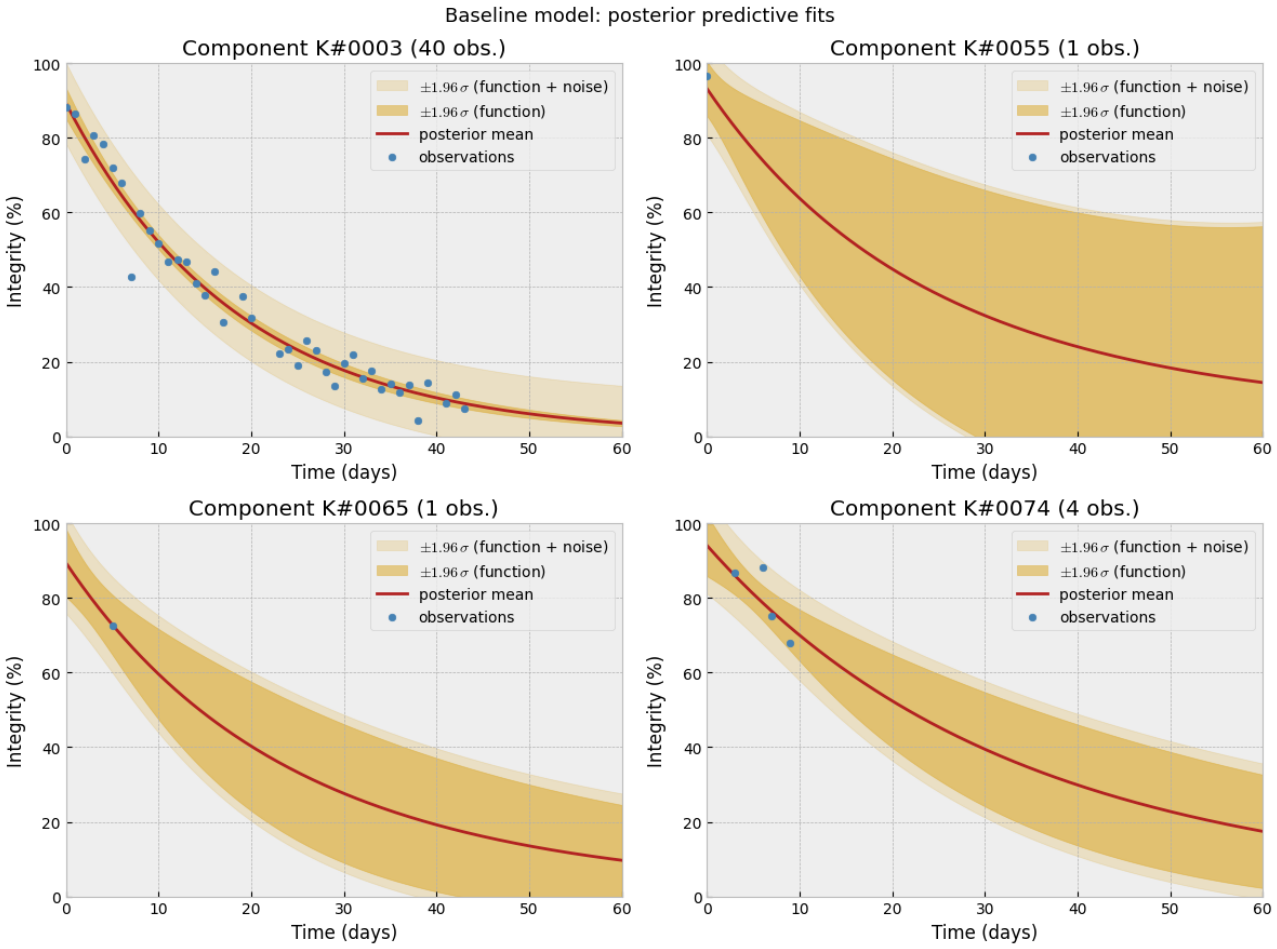


Figure 2. Baseline model posterior predictive fits: posterior mean predictor and uncertainty bands. Inner uncertainty band captures 95% of function $f_i(t)$ uncertainty. Outer uncertainty band captures 95% of function $f_i(t)$ and measurement noise σ_y uncertainty together.

3.2 Enhanced Model

The enhanced model keeps the population level structure of the baseline but more accurately represents the decay rate by accounting for the five diagnostics. The across component spread of the decay rate v_i drops from σ_v = 1.59 (baseline) to σ_v = 0.35 (enhanced): a 78% reduction in standard deviation. The observation noise σ_y is essentially unchanged (5.10 to 5.09) as is expected: the diagnostics explain variation between components but cannot reduce shared component measurement noise. Looking at Figure 3, similar observations about unimodal, not heavily skewed posterior distributions with indistinguishable chains can be made. The means μ_v and μ_u have changed marginally (3.93 to 4.17, 89.72 to 90.33) indicating a likely marginally better population mean estimate. Posterior mean σ_u also drops slightly from 5.28 to 4.78 indicating a slight improvement in across component spread of the intercept estimate. Estimates remain within the client's suggested $u_i \in [80, 100]$, $v_i \in [1, 10]$).

The most valuable consequence of incorporating the diagnostics is visible in the predictions shown in Figure 4. Compared to Figure 2, the function-uncertainty bands are much tighter for the components with limited data. For K#0055 the 95% uncertainty band at t = 30 narrows from roughly [0%, 70%] under the baseline model to approximately [10%, 40%] under the enhanced model and the posterior mean predictor is noticeably different. This means much more confident predictions for components with limited data. Predictions for K#0003 with lots of data changes minimally.

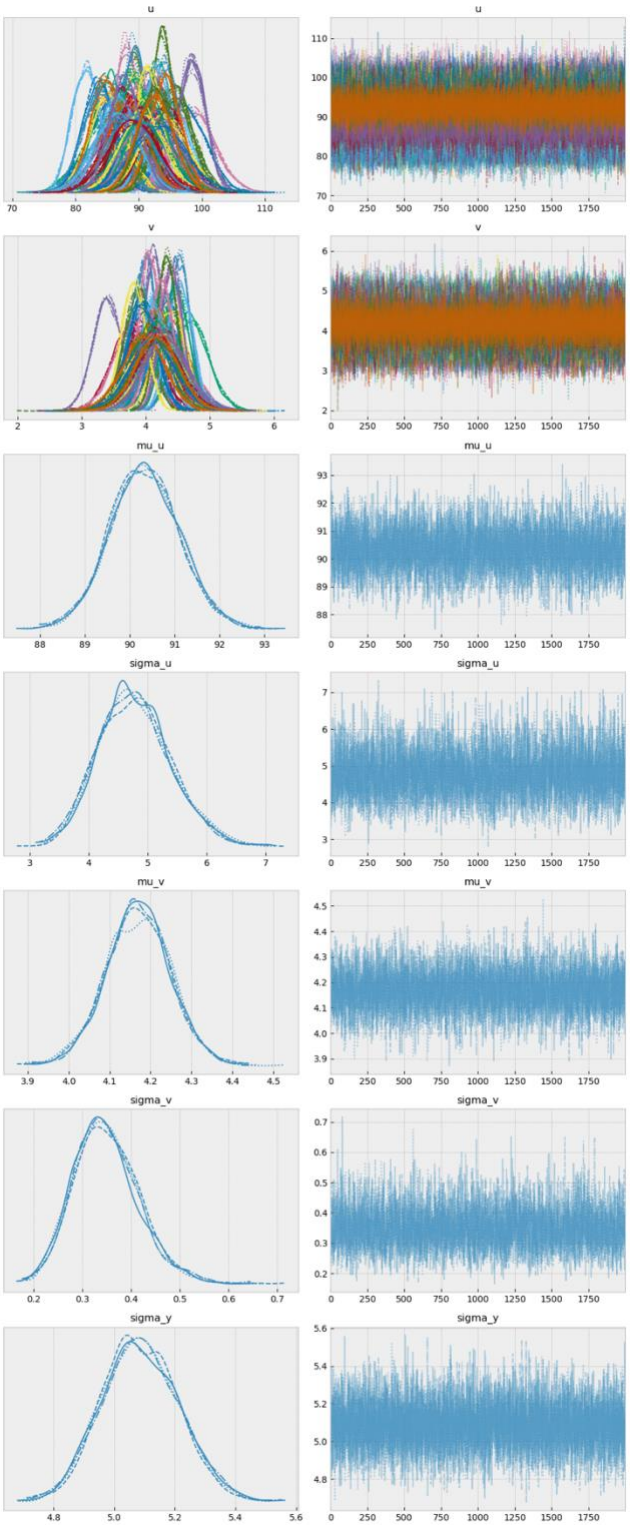


Figure 3. Enhanced model posterior diagnostics. Posterior distributions, one curve per chain, and trace of sampled values for component parameters and measurement noise. The first two rows show all 75 per component u_i and v_i on one graph.

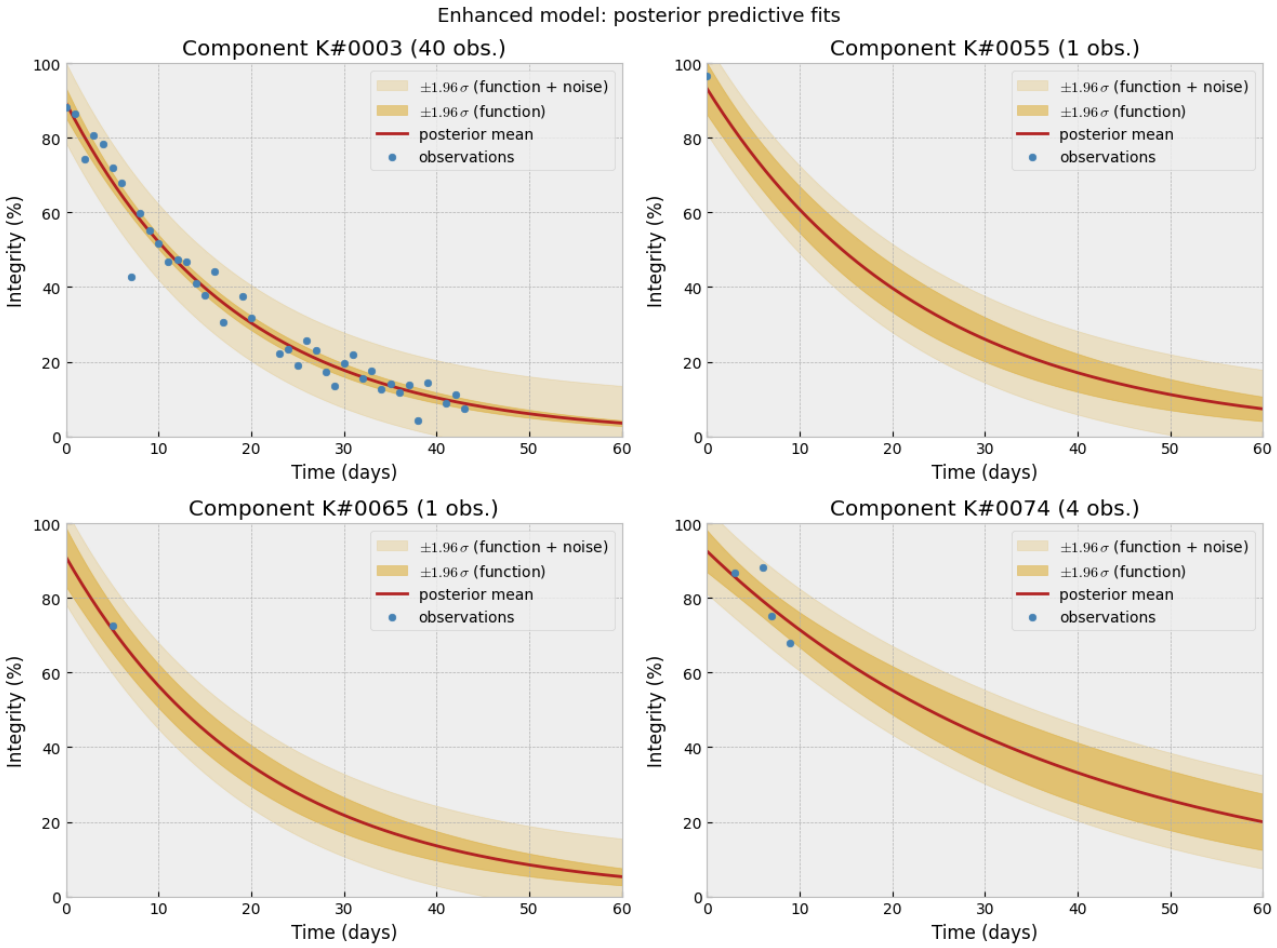


Figure 4. Enhanced model posterior predictive fits: posterior mean predictor and 95% function and measurement noise uncertainty bands.

4. Potential Causes of Efficiency Loss

The posterior distributions of the diagnostic weights w_j and their shared scale σ_w are shown in Figures 5 and 6. Recall that the diagnostics were standardised before fitting and so the magnitudes of the w_j are directly comparable. Table 1 also provides the equivalent weights on the unstandardised X scale.

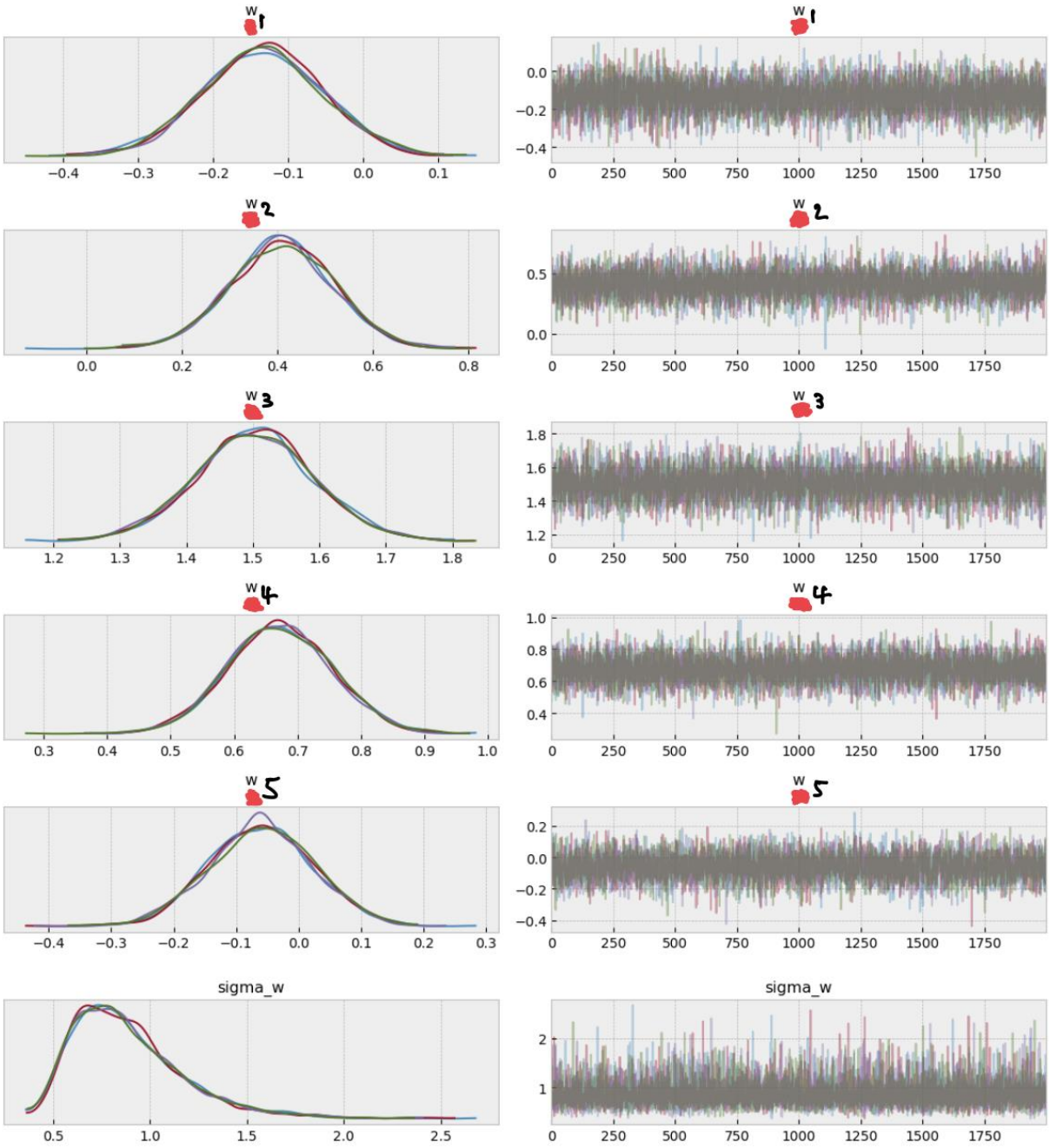


Figure 5. Enhanced model trace and density plots for the diagnostic weights $w_1, \ldots, w_5$ (corresponding to X1,…, X5) and shared scale σ_w. w_2, w_3 and w_4 are positive and so higher values of the diagnostics X2, X3 and X4 result in increased rate of component K degradation. w_5 is centred around 0 (no effect) and w_1 is slightly negative.

Using the data in Table 1, conducting a Bayesian hypothesis test to test if each weight is non zero (Statistics LibreTexts, 2019) gives a marginal prior at w=0 which diverges meaning that the Bayes factor for the null H_0: $w_i = 0$ is not well defined. Being informally Bayesian, we use the 95% credible intervals of the posterior weights to check if a weight of 0 (no effect on integrity) is likely (Clyde, 2021). The credible intervals for w_3, w_4 and w_2 all do not include 0 and so there is evidence that these weights are non-zero. The credible intervals for w_1 and w_5 include 0 so there is

insufficient evident to suggest these weights are non-zero. From this, we conclude that diagnostics X3, X4 and X2 influence degradation.

Table 1. Posterior summary of diagnostic weights.

Weight	Posterior mean (standardised X)	95% CI (std X)	P(w > 0)	Mean (original X)
w_1	−0.134	[−0.294, +0.023]	0.05	−0.086
w_2	+0.409	[+0.195, +0.620]	≈1.00	+0.168
w_3	+1.503	[+1.323, +1.683]	≈1.00	+1.066
w_4	+0.672	[+0.513, +0.832]	≈1.00	+0.225
w_5	−0.060	[−0.219, +0.101]	0.24	−0.030

Key finding: The diagnostic **X3 is by far the strongest predictor of degradation (+1.503)**. X4 is the next most important (+0.672), and X2 is also moderately strong (+0.409). As explained above, there is insufficient evidence to suggest that X1 and X5 influence degradation rate. **Intervention should be focused on diagnostics X3, X4 and X2.** Intervention directed at X1 or X5 is unlikely to produce any benefit.

5. Blind Test Predictions

The 25 blind test probabilities - the posterior predictive probability that each test component's integrity is 30% or below at day 30 – were then computed under the enhanced model. This was calculated by counting the number of sampled functions f that predict $y \leq 30$ at $t = 30$ and dividing by the total number of sampled functions. The summary distribution of probabilities is shown in Figure 7.

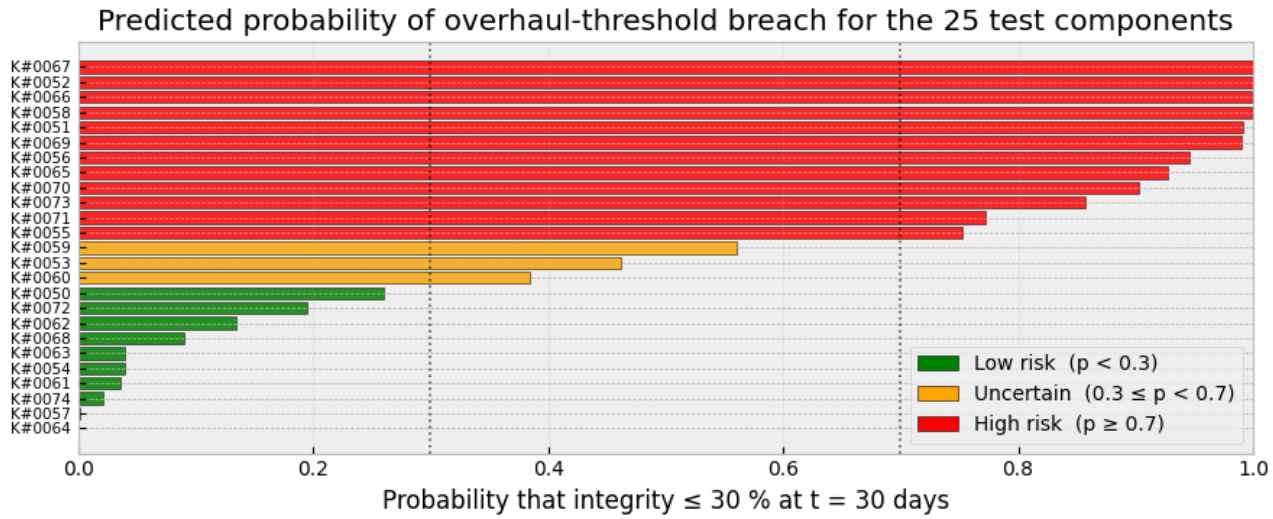


Figure 7. Blind test probabilities that component integrity is at or below 30% at day 30. The dotted lines mark suggested thresholds at 0.3 and 0.7 — components in green are likely to be safe at 30 days, those in orange are uncertain and should be checked for performance and those in red require replacement.

Of the 25 test components, 12 have a probability above 0.7 of being above the intervention threshold by day 30, 10 have a probability below 0.3 and are likely safe; and 3 are in the uncertain range where additional checks would provide the most value.

6. Executive Evaluation

6.1 Why use a Hierarchical Bayesian Approach?

Standard supervised machine learning approaches – linear regression, decision trees, neural networks - produce point predictions with usually no measure of uncertainty and are typically fitted by minimising an average loss function over the entire training set. This produces predictions that are shifted toward the population average. These methods and frequentist statistics in general, ignore uncertainty in the value of parameters, giving predictive intervals that are overconfident (Lambert, 2018). Quantifying uncertainty accurately is particularly important for predicting the future degradation of the 25 components with little data and so a hierarchical Bayesian approach is extremely useful in this setting.

Hierarchical Bayesian modelling allows information to spread from components with lots of data to components with limited data with its uncertainty adjusted to reflect that lack of data. We fit two different models: a baseline model and an enhanced model incorporating diagnostic information into the decay rate of each component. Both models were fitted with the HMC sampler and NUTS kernel. Informative priors were chosen for each parameter to encode prior knowledge. The relative importance of each diagnostic was then found and predictions produced for the sparse data components.

6.2 Strengths of the Method

Uncertainty: the wider error bars on K#0055 in Figure 4 indicate that with one observation at t ≈ 0, the future degradation is reasonably uncertain. A point prediction model might give the client false confidence in their prediction or understate the uncertainty.

Predicting new data: providing predictions for just 25 components is computationally inexpensive and quick because of the vectorised predict function. Refer to Section 6.3 for limitations when predicting vastly more data.

Interpretability and domain knowledge: domain expertise is incorporated through the exponential decay form of $f_i(t)$ and the parameters u_i, v_i and w_j are directly interpretable. This is generally not possible with a black box predictor like a neural network.

6.3 Limitations

Computational cost: HMC is fast for this problem (around 10 seconds per fit on a four core CPU) but is not free and scales with number of parameters and amount of training data. Inference for many, many components would require a change in architecture such as variational inference (see Section 6.4).

Model Drift: the diagnostics X1–X5 are constant across time per component; if these values change over time, the model will not capture that drift and predictions will become inaccurate.

6.4 Recommended Next Steps

Investigate X3: X3 very strongly predicts component K degradation: the client should investigate whether X3 is controllable and if so, whether interventions targeting X3 can extend service life and prevent replacement.

Plan for scaling: if deployment using much more data is carried out, variational inference for the same model should be explored. This which would reduce the computational cost massively with a small accuracy loss.

7. Conclusions

This report presents a Bayesian hierarchical predictive model of component K integrity. The Bayesian framework detailed here is, in my judgement, the most informative method available for this problem. It has given insights into targeting X3, X4 and X2 in any intervention measures and provides informative predictions for the degradation of components with limited data: 12 of 25 test components have predicted probability more than 0.7 of exceeding the 30% integrity threshold by day 30, 3 test components should be checked for maintenance and 10 have probability less than 0.3 of exceeding this threshold.

8. References

Clyde, M. (2021). *Lecture 5: Basics of Bayesian Hypothesis Testing*. [online] Available at: https://www2.stat.duke.edu/courses/Fall21/sta601.001/slides/05-hypothesis-testing-handout.pdf [Accessed 8 May 2026].

Lambert, B. (2018). *A Student's Guide to Bayesian Statistics*. SAGE.

Statistics LibreTexts. (2019). *17.2: Bayesian Hypothesis Tests*. [online] Available at: https://stats.libretexts.org/Bookshelves/Applied_Statistics/Learning_Statistics_with_R_-_A_tutorial_for_Psychology_Students_and_other_Beginners_(Navarro)/17%3A_Bayesian_Statistics/17.02%3A_Bayesian_Hypothesis_Tests.